METHODS, SYSTEMS, AND APPARATUS FOR ARTIFICIAL GENERATION, CONTROL, AND APPLICATION OF SCALAR DENSITY FIELDS

Inventor: Gary Thomas Alcock

Applicant: Gary Thomas Alcock

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Abstract

Disclosed are methods, systems, and apparatus for artificial generation, control, and application of a scalar density field that simultaneously governs electromagnetic propagation and matter acceleration. In embodiments, a controller shapes a commanded field within a chamber using configurable sources, measures the resulting optical and inertial responses, and closes feedback to achieve a desired field profile. The technology enables propulsion, refractive steering, wireless energy transfer, precision metrology, navigation and timing, and calibration of line-of-sight environmental biases.

Cross-Reference to Related Applications

This application claims the benefit of any U.S. provisional application filed by the Applicant on or after August 15th, 2025, the entirety of which is incorporated herein by reference.

Field of the Invention

The disclosure relates to physics, metrology, and engineered systems for artificially generating, manipulating, and using a scalar density field to control both electromagnetic propagation and matter acceleration, including laboratory, aerospace, and geophysical applications.

Background

Conventional technologies separately control light (via optical elements and media) and inertial motion (via masses or spacetime curvature). No unified platform provides tunable control over both photon trajectories and effective gravitational responses using the same field mechanism and feedback architecture. There is a need for practical systems to generate, measure, stabilize, and apply a scalar field that couples to both domains.

Summary

In embodiments, a controller generates an artificial scalar field within a chamber by arranging matter and/or engineered media and driving actuators to realize a target field profile. The resulting field alters electromagnetic propagation via a refractive response and induces matter acceleration via field gradients. Sensing (e.g., optical/atomic interferometry and accelerometry) provides measurements for closed-loop control, safety enforcement, and application-specific operation. Representative applications include propulsion, refractive beamforming, wireless energy transfer, navigation and timing, materials processing, and environmental bias calibration.

Brief Description of the Drawings

- **FIG. 1** is a schematic of an artificial field generator including a chamber, configurable sources, environmental controls, sensors, and control electronics.
- **FIG. 2** is a functional block diagram of closed-loop field synthesis with estimation and safety limits.
- **FIG. 3** illustrates representative optical and inertial couplings: refractive index as a function of the scalar field and induced acceleration as a function of field gradient.

- **FIG. 4** is a laboratory interferometer configuration enabling precision measurement of field-induced path changes.
- **FIG. 5** depicts exemplary applications including propulsion by directed field gradients and wireless energy transfer by refractive steering.

Detailed Description

System Overview (FIGS. 1–2)

In one embodiment, an apparatus includes (i) a field generation chamber housing configurable sources; (ii) environmental controls to set pressure and temperature; (iii) a sensor suite including accelerometers, optical/atomic interferometers, and timing links; and (iv) control electronics implementing real-time estimation, feedback, and safety enforcement. FIG. 1 shows one example layout. FIG. 2 depicts a feedback architecture wherein a target field is computed, actuation commands are synthesized for multiple zones, measurements are fused into a field estimate, and constraints enforce operating limits.

Refractive and Inertial Responses (FIG. 3)

In embodiments, light propagation experiences an effective refractive response to the scalar field, and matter experiences an effective acceleration response to the field gradient. FIG. 3 provides representative relationships for design and calibration. These responses enable refractive beamforming, guided paths, and effective force generation or nulling.

Laboratory Measurement (FIG. 4)

FIG. 4 shows an interferometric measurement suitable for precision mapping of field-induced path differences. Two separated arms provide sensitivity to vertical or horizontal gradients. The configuration is compatible with optical frequency combs, cavities, and fiber links for timing and phase readout.

Applications (FIG. 5)

Directed field gradients produce net effective acceleration on structures for propulsion or levitation. Refractive steering forms corridors for energy or signal transfer between terminals. Additional applications include navigation/time transfer via route-dependent path delays, materials processing via controlled gradients, and calibration of environmental line-of-sight biases in precision measurements.

Safety and Control

Hardware and software impose bounds on field magnitude, gradient, and rate of change. Monitors compare real-time estimates against thresholds, and violations trigger quench or safe-state reversion. Multi-zone control enables spatial shaping while preserving safety margins.

Enablement Notes

Examples include static and dynamic profiles, single-zone and multi-zone sources, and multiple sensor modalities. Actuation can employ magnetic, electrostatic, acoustic, thermal, mechanical, or engineered-media methods. The invention is implementation-agnostic so long as the system generates, measures, and uses a scalar field with optical and inertial effects as described.

FIG. 1 — Artificial field generator (schematic).

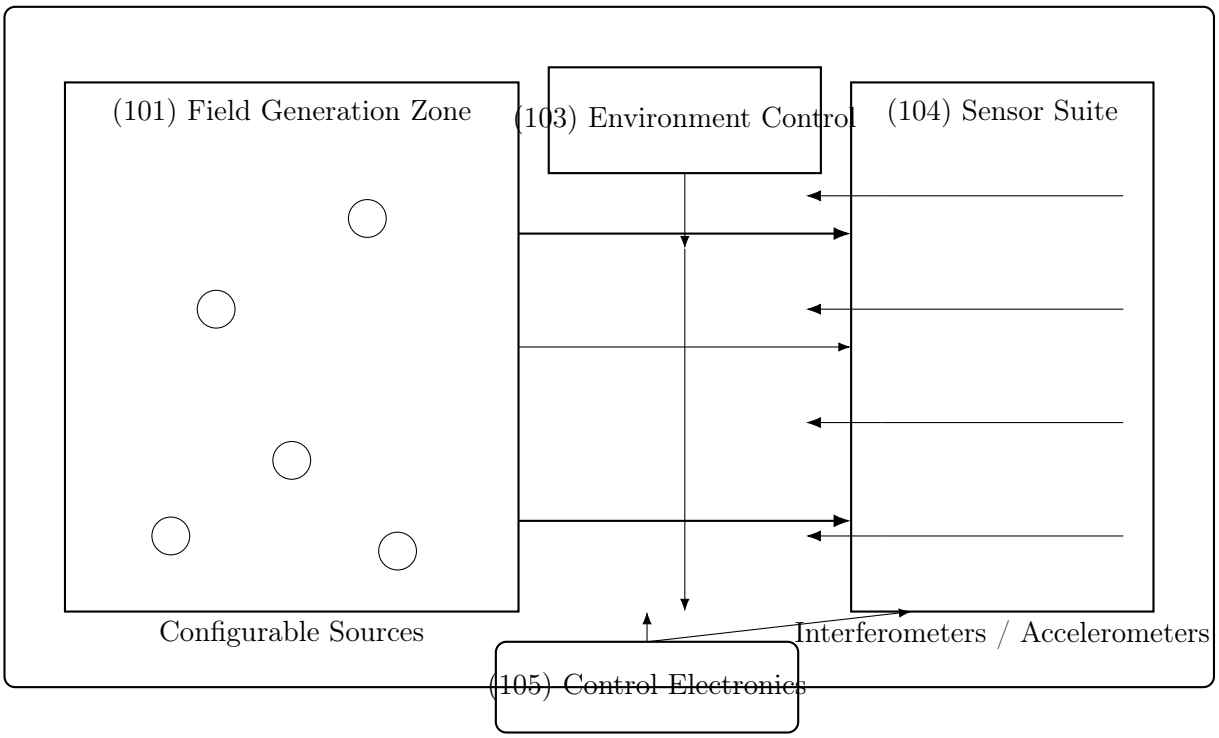


Figure 1: *

Apparatus includes a chamber with configurable sources (101), environmental controls (103), sensors (104), and control electronics (105).

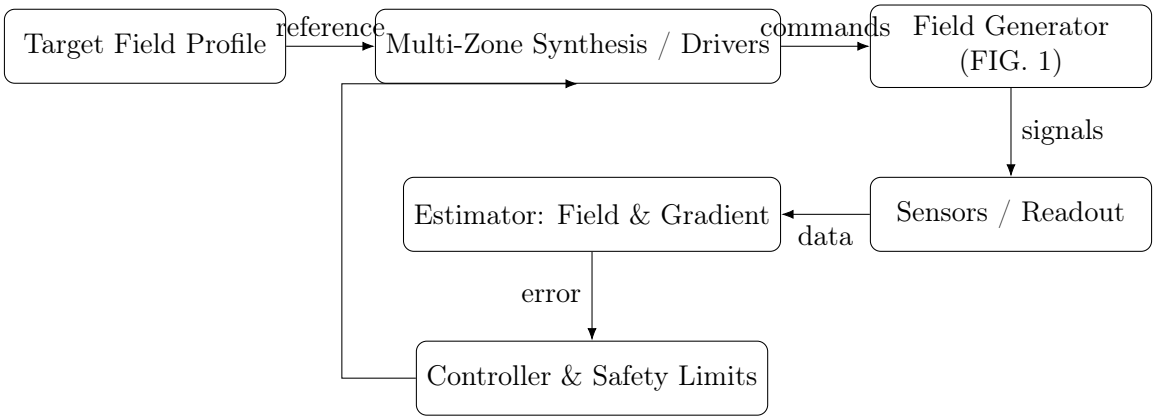


Figure 2: Closed-loop field synthesis with estimation and safety. Feedback drives the measured field toward the target while enforcing bounds on magnitude, gradient, and slew rate.

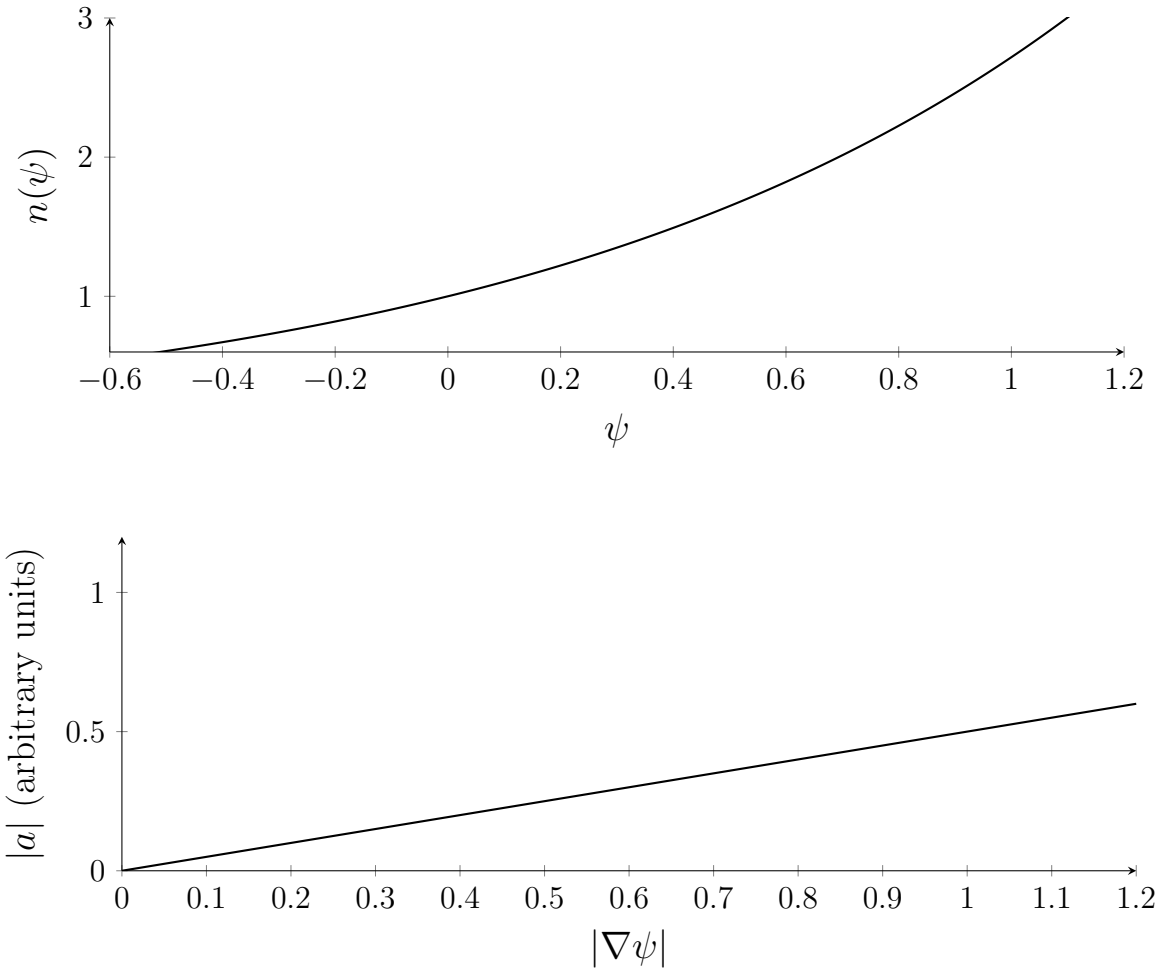


Figure 3: *

FIG. 3 — Representative couplings: refractive response $n(\psi)$ (top) and induced acceleration vs. gradient (bottom).

FIG. 4 — Interferometer configuration for precision path measurement.

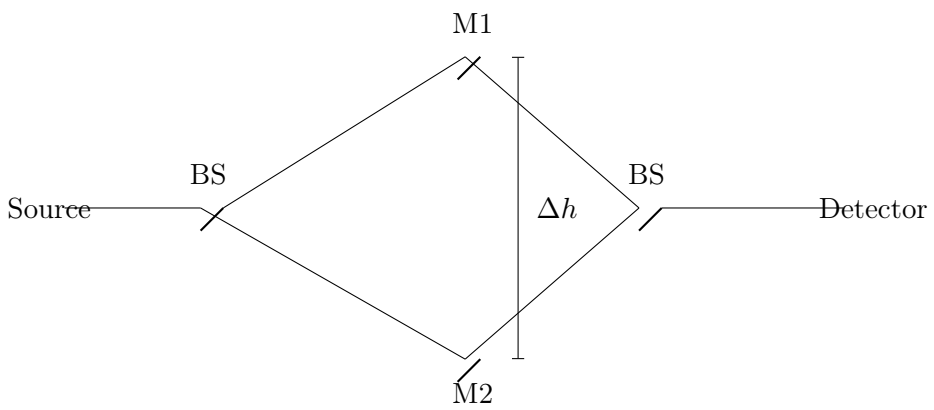
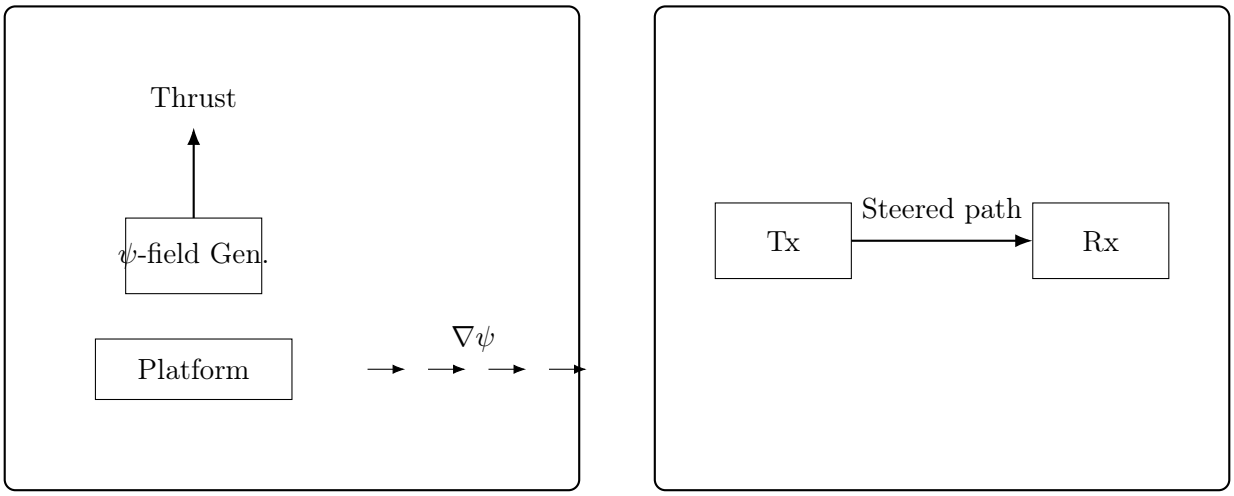


Figure 4: *

Vertical separation Δh provides sensitivity to field gradients; compatible with cavity/comb readouts.

FIG. 5 — Representative applications.



(a) Propulsion by directed gradients

(b) Wireless energy via refractive steering

Figure 5: *

(a) Directed field gradients impart net acceleration; (b) Refractive steering forms a guided energy/signal corridor.

Claims

1. **(Independent)** A method of inducing acceleration and controlling electromagnetic propagation, comprising: generating, within a chamber, an artificial scalar field that is responsive to at least one configurable source; and operating a controller to shape a target field profile while measuring optical and inertial responses, whereby electromagnetic propagation and matter acceleration are controlled using a common field within said chamber.

2. The method of claim 1, wherein electromagnetic propagation is controlled by a refractive response characterized by a field-dependent index.

3. The method of claim 1, wherein matter acceleration is controlled by a gradient response characterized by an induced acceleration proportional to a field gradient.

4. The method of claim 1, further comprising estimating a field state and closing feedback to reduce error between a measured field and said target field profile.

5. The method of claim 1, further comprising enforcing safety limits on at least one of field magnitude, gradient magnitude, spatial extent, and temporal rate of change.

6. The method of claim 1, wherein shaping the target field profile comprises multi-zone synthesis using a plurality of independently driven source zones.

7. The method of claim 1, wherein the method is applied to at least one of: laboratory interferometry, propulsion or levitation of a structure, wireless energy or signal transfer by refractive steering, navigation or timing via route-dependent path delays, materials processing, or environmental bias calibration.

8. The method of claim 1, wherein configurable sources comprise at least one of: mass positioning, engineered media, magnetic, electrostatic, acoustic, thermal, mechanical, or hybrid actuation.

9. The method of claim 1, wherein sensors comprise at least one of: optical interferometers, atomic interferometers, accelerometers, or timing links.

10. **(Independent)** A system for controlling electromagnetic propagation and matter acceleration through artificial scalar field generation, comprising: a chamber containing configurable sources; a sensor suite configured to measure optical and inertial responses within the chamber; and control electronics configured to synthesize a target field profile and enforce operating constraints, wherein multiple applications are implemented using the same field control technology, including at least one of: astrophysical or cosmological modeling, laboratory experimentation, refractive steering for energy or communication, propulsion or levitation, or environmental bias calibration.

11. The system of claim 10, wherein the configurable sources comprise a plurality of independently actuated zones for spatial field shaping.

12. The system of claim 10, wherein the sensor suite includes an interferometer with separated arms providing sensitivity to vertical or horizontal gradients.

13. The system of claim 10, wherein the control electronics comprise a real-time estimator and controller operative to reduce error between a measured field and a commanded field.

14. The system of claim 10, wherein the control electronics are configured to automatically quench sources upon detection of a threshold exceedance of at least one of field magnitude, gradient, or slew rate.

15. The system of claim 10, further comprising environmental controls to set pressure and temperature within the chamber for stable operation.